SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 — Precision	**Assessment:**	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
Student Name:		**Reg. No.:**	
Section / Batch:		**Date:**	

Code of Conduct

I ___________________________(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __________________________

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Frame Semantics in Banking Customer-Service Systems

A digital banking platform uses semantic frames to understand customer requests. The NLP system identifies the **action, agent, object, and destination** involved in each request.

Customer Query	Generated Semantic Frame
"Transfer ₹20,000 from my savings account to my son's account."	TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)
"Block my debit card immediately."	BLOCK(DebitCard, Customer)
"Send my transaction statement to my email."	SEND(TransactionStatement, Email, Customer)
"Increase my daily withdrawal limit."	MODIFY(WithdrawalLimit, Customer)

Additional transaction logs show:

Query ID	Actual User Intent	System Interpretation
B1	Transfer money to another account	Transfer money
B2	Block debit card	Block debit card
B3	Receive transaction statement by email	Send statement to email
B4	Increase withdrawal limit	Decrease withdrawal limit

Tasks:

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.
2. Identify the query in which the semantic interpretation is incorrect and explain why.
3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.
4. Recommend a frame-based semantic analysis approach to improve the reliability of banking conversational systems.

Use intent detection, semantic-role/slot labeling, banking-rule validation, and confirmation for high-risk operations.

4. Recommended approach:

Incorrect semantic roles can cause wrong banking operations, such as reducing a limit instead of increasing it or using an incorrect transfer destination.

3. Effect:

B4 is incorrect. The user wants to increase the withdrawal limit, but the system interpreted it as decrease.

2. Incorrect interpretation:

• Modify → WithdrawalLimit = object, Customer = agent.

• Send → TransactionStatement = object, Email = destination, Customer = agent.

• Block → DebitCard = object, Customer = agent.

• Transfer → SourceAccount = savings account, Amount = ₹20,000, DestinationAccount = son's account, Customer = customer.

1. Semantic frames:

Answer

FRAME SEMANTICS IN BANKING CUSTOMER-SERVICE SYSTEMS

1.1 Semantic Frame Analysis

Each banking request maps to semantic frames with specific roles:

Transfer: TRANSFER(Agent=Customer, Theme=Money, Destination=Account)
Block card: BLOCK(Agent=Customer, Patient=Card, Goal=BlockedState)
Send statement: SEND(Agent=System, Theme=Statement, Goal=Email)
Increase limit: MODIFY(Agent=Customer, Patient=Limit, Direction=increase, Magnitude=quantifier)

Semantic Relationship Pattern: Customer is typically the Agent (entity whose request triggers action), while the Object (card, limit, account) is the Patient (entity undergoing modification).

1.2 Incorrect Interpretation Identified

Query B4 Contains an ERROR:

Actual Intent: "Increase my daily withdrawal limit"
System Interpretation: "Decrease withdrawal limit" ✗ INCORRECT

Root Cause: The system failed to recognize the directional modifier "increase" and inverted the semantic operation to its opposite.

1.3 Impact of Incorrect Role Identification

Consequences:

1. Wrong Operation: Customer's limit would be DECREASED not INCREASED
2. Financial Harm: Customer locked out of requested transactions
3. Regulatory Risk: Violates customer's explicit request; compliance violation
4. Trust Erosion: Banking system credibility damaged

Banking operations must be reversible and verifiable. Executing the opposite operation causes material financial and legal harm.

1.4 Recommended Frame-Based Semantic Architecture

Proposed Improvements:

Multi-Level Parsing:
• Syntactic parsing to identify directional modifiers (increase/decrease)
• Negation detection to distinguish "don't increase" from "increase"
• Temporal markers to identify urgency ("immediately")

Semantic Role Labeling (SRL) Framework:
• Use Prop Bank/Frame Net with banking-specific frame extensions
• Define BANKING frames with explicit role sets

Confidence & Verification:
• Score confidence; flag low-confidence cases (< 0.85) for human review
• Require explicit confirmation: "I will INCREASE your limit from X to Y. Correct?"
• Bidirectional consistency checks against customer preferences

Error Prevention & Audit Trail:
• Validate reverse operation to ensure idem potency
• Immutable transaction logging with frame structure and confidence scores

2. First-Order Predicate Calculus for Smart Agriculture

A smart agricultural system monitors crop fields using sensors and logical rules.

Field Data:

Field	**Soil Condition**	**Irrigation Status**	**Crop**
F1	Dry	Available	Rice
F2	Wet	Available	Wheat
F3	Dry	Unavailable	Maize
F4	Wet	Unavailable	Rice

The system uses the following predicate rules:

- Dry(x) ∧ IrrigationAvailable(x) → NeedsWater(x)
- NeedsWater(x) → Irrigate(x)

- Wet(x) → ¬NeedsWater(x)
- IrrigationUnavailable(x) → ¬Irrigate(x)
- Rice(x) ∧ NeedsWater(x) → PriorityIrrigation(x)

Tasks:

1. Represent the field observations using First-Order Predicate Calculus.
2. Using the given rules, determine which fields require irrigation.
3. Determine whether F1 and F3 should be treated differently by the automated irrigation controller.
4. Analyze whether predicate logic alone is sufficient for agricultural decision support when sensor information is incomplete or uncertain.

Predicate logic handles deterministic rules, but incomplete or uncertain sensor data requires probabilistic reasoning, fuzzy logic, confidence scores, or sensor fusion.

4. Predicate logic:

F1 should be irrigated because water is available. F3 should not be irrigated because irrigation is unavailable.

3. F1 vs F3:

F1 → NeedsWater(F1) → Irrigate(F1) and PriorityIrrigation(F1). F2 and F4 are wet, so they do not need water. F3 is dry but irrigation is unavailable. Therefore, F1 requires irrigation.

2. Irrigation:

F4: Wet(F4) ∧ IrrigationUnavailable(F4) ∧ Rice(F4) F3:

Dry(F3) ∧ IrrigationUnavailable(F3) ∧ Maize(F3) F2:

Wet(F2) ∧ IrrigationAvailable(F2) ∧ Wheat(F2) F1:

Dry(F1) ∧ IrrigationAvailable(F1) ∧ Rice(F1)

1. FOPC:

Answer

FIRST-ORDER PREDICATE CALCULUS FOR SMART AGRICULTURE

2.1 Field Observations in Predicate Calculus

Formal representations:

F1: Dry(F1) ∧ IrrigationAvailable(F1) ∧ Rice(F1)

F2: Wet(F2) ∧ IrrigationAvailable(F2) ∧ Wheat(F2)

F3: Dry(F3) ∧ IrrigationUnavailable(F3) ∧ Maize(F3)

F4: Wet(F4) ∧ IrrigationUnavailable(F4) ∧ Rice(F4)

2.2 Fields Requiring Irrigation

Applying predicate rules:

F1: Dry(F1) ∧ IrrigationAvailable(F1) → NeedsWater(F1) → Irrigate(F1) ✓ IRRIGATE
F2: Wet(F2) → ¬NeedsWater(F2) ✗ No irrigation
F3: Dry(F3) ∧ IrrigationUnavailable(F3) → ¬Irrigate(F3) ✗ Cannot irrigate
F4: Wet(F4) → ¬NeedsWater(F4) ✗ No irrigation

ANSWER: Only Field F1 requires irrigation.

2.3 Differential Treatment of F1 vs. F3

Despite both having dry soil:

F1 (Dry + Available): Can be resolved through irrigation → IRRIGATE immediately
F3 (Dry + Unavailable): Represents constraint violation → ALERT: escalate to alternative strategy (drought-resistant crops, water sourcing)

Critical Distinction: F1 is a solvable problem (need exists, resource available). F3 is a constraint violation (need exists, resource unavailable). F3 requires meta-level reasoning and resource reallocation.

2.4 Limitations of Predicate Logic Alone

Key Limitations:

1. Binary Truth Values: Soil moisture is continuous, not binary (Dry or ¬Dry)
2. Incomplete Information Assumption: Sensors have errors; weather unpredictable
3. No Uncertainty Representation: Sensor malfunction, equipment failure unmodeled
4. No Cost-Benefit Analysis: Economic factors (costs, water scarcity pricing) ignored
5. No Temporal Reasoning: Growth stage, seasonal patterns, forecasts not modeled

Recommended Enhancements:

• Fuzzy Logic: Represent soil moisture as membership degrees (0.0-1.0)
• Probabilistic Reasoning: Bayesian networks for sensor uncertainty
• Machine Learning: Learn patterns of crop needs vs. sensor data
• Constraint Optimization: Handle resource constraints with mixed-integer programming
• Temporal Reasoning: Time-aware rules capturing growth stage and forecasts

3. Word Sense Disambiguation in a Travel Recommendation System

A travel platform receives search queries containing ambiguous words. The recommendation engine uses surrounding words and user interactions to determine the intended meaning.

Search Query	Possible Sense 1	Possible Sense 2
"Amazon tour"	Amazon rainforest	Amazon company
"Safari booking"	Wildlife tour	Web browser
"Java vacation"	Java island	Java programming language
"Cruise package"	Ship journey	Cruise software/project

The system records the following user interactions:

Query	User Interaction
Amazon tour	Viewed rainforest trekking packages
Safari booking	Selected wildlife resort packages
Java vacation	Viewed hotels in Bali and Java
Cruise package	Viewed Mediterranean ship packages

However, another user searches: **"Safari download for Windows"** and clicks a browser download page.

Tasks:

1. Determine the intended sense of the ambiguous terms using the available context.
2. Identify the lexical and contextual cues that distinguish the possible senses.
3. Analyze why the same word may require different interpretations for different users.
4. Design an industrial-scale WSD strategy using query context, user history, embeddings, and click behaviour to improve travel recommendations.

Combine query context, user history, embeddings, and click behaviour to generate, score, and rank candidate senses. Use a confidence threshold when clarification is needed.

4. WSD strategy:

The same word can have different meanings depending on user goals, history, surrounding words, and clicks.

3. Different users:

Tour, trekking, wildlife, vacation, hotels, and ship indicate travel senses. Download and Windows indicate the browser/software sense.

2. Cues:

• Safari download for Windows → Web browser.

• Cruise package → Ship journey.

• Java vacation → Java island.

• Safari booking → Wildlife tour.

• Amazon tour → Amazon rainforest.

1. Intended senses:

Answer

3. WORD SENSE DISAMBIGUATION IN TRAVEL RECOMMENDATIONS

3.1 Intended Sense Determination

"Amazon tour" → Rainforest sense (user viewed rainforest trekking packages) ✓
"Safari booking" → Wildlife tour sense (user selected wildlife resort packages) ✓
"Java vacation" → Island sense (user viewed hotels in Java, geographic context) ✓
"Cruise package" → Ship journey sense (user viewed Mediterranean cruises) ✓
"Safari download Windows" → Browser sense (user clicked .exe download page; pattern "X download for OS" indicates software) ✓

3.2 Lexical and Contextual Cues

Lexical Cues (Word Co-occurrence):

- "tour", "booking", "vacation", "package" → travel domain signals
- "download for Windows" → software download pattern

Contextual Cues (User Behavior):

- Click destinations: rainforest trekking, wildlife resorts affirm travel sense
- Dwell time on results indicates sense preference
- Search history (flights, hotels) establishes user profile bias
- Session context: travel planning vs tech setup

Domain Signals (Platform Context):

- Travel platform provides initial bias toward travel senses
- "download" verb typically redirects to software, overriding platform bias

3.3 Why Same Word Requires Different Interpretations

Example: "Safari" requires different interpretations in different contexts

Context 1 - "Safari booking":
→ Query: "booking" is travel verb
→ Behavior: Wildlife resort selection
→ Sense: Wildlife tour

Context 2 - "Safari download Windows":
→ Query: "download for OS" is software pattern
→ Action: File download (.exe) page click
→ Sense: Web browser

Factors Requiring Differentiation:

1. Query structure activates different semantic fields
2. User intent signals (resort selection vs file download)
3. Session context (travel planning vs tech setup)
4. Temporal signals (holiday season vs OS update)
5. User profile (travel history vs tech habits)

3.4 Industrial-Scale WSD Strategy

Multi-Signal Ensemble Architecture:

LAYER 1: Query Context Analysis
• POS tagging & dependency parsing
• Named entity recognition (geographic, companies)
• Semantic field detection (e.g., "download + Windows" → software)

LAYER 2: Word Embeddings (30% weight)
• Contextual embeddings (BERT, RoBERTa)
• Compare vectors to sense-specific embeddings
• Cosine similarity scoring

LAYER 3: User History & Profile (20% weight)
• Tech vs travel query ratio
• Previous sense choices in ambiguous queries
• Device/OS preferences

LAYER 4: Click Behavior (20% weight)
• Landing page type (travel booking vs app download)
• Dwell time on results
• Follow-up queries and conversion signals

LAYER 5: Ensemble Scoring
Score = 0.30×context + 0.30×embedding + 0.20×history + 0.20×click
Confidence > 0.85: Recommend primary sense
Confidence 0.60-0.85: Show both senses; track user choice
Confidence < 0.60: Ask user for clarification

Key Techniques:

- Contextual embeddings capture surrounding context meaning
- Knowledge graphs link senses to encyclopedia entities
- Session-level analysis tracks discourse context
- Feedback loops: Log clicks; retrain model with user feedback

4. Syntax-Driven Semantic Analysis in Legal Document Processing

A legal NLP system analyzes sentences from contracts and assigns semantic roles based on their syntactic structures. The system produces the following analyses:

Legal Sentence	Parse Structure	Generated Semantic Interpretation
"The supplier delivered the equipment to the buyer."	Subject–Verb–Object	Supplier = Agent, Equipment = Object, Buyer = Recipient
"The buyer received the equipment from the supplier."	Subject–Verb–Object	Buyer = Agent, Equipment = Object, Supplier = Recipient
"The company terminated the contract after the violation."	Subject–Verb–Object	Company = Agent, Contract = Object
"The contract was terminated by the company."	Passive Construction	Contract = Agent, Company = Object

Tasks:

1. Analyze how syntactic structure, including active and passive constructions, influences semantic-role assignment.
2. Identify the incorrect semantic-role assignments in the given interpretations.
3. Explain how confusing grammatical subject with semantic agent can affect automated legal information extraction.
4. Recommend a syntax-driven semantic analysis architecture that can correctly handle active voice, passive voice, prepositional phrases, and long-distance dependencies in legal documents.

Use dependency/constituency parsing, voice detection, semantic-role labeling, prepositional-phrase analysis, coreference resolution, and consistency checking.

4. Recommended architecture:

Confusing grammatical subject with semantic Agent can reverse legal responsibilities and actions during information extraction.

3. Effect:

Sentence 4: Contract = Agent and Company = Object are incorrect. Contract is the affected entity and Company is the Agent.

Sentence 2: Supplier = Recipient is incorrect; "from the supplier" indicates the source.

2. Incorrect assignments:

In active voice, the subject commonly acts as Agent and the object as the affected entity. In passive voice, the subject is usually the affected entity and the Agent may appear in a "by" phrase.

1. Syntactic influence:

ANSWER

SYNTAX-DRIVEN SEMANTIC ANALYSIS IN LEGAL DOCUMENTS

4.1 Syntactic Structure Influence on Semantic Roles

Active Construction: "The supplier delivered the equipment"
Subject → Supplier | Agent=Supplier, Patient=Equipment

Passive Construction: "The equipment was delivered by the supplier"
Subject → Equipment | Agent=Supplier (by-phrase), Patient=Equipment

CRITICAL: Grammatical subject ≠ Semantic agent in passive voice

Prepositional Phrase Signals:
- "by" agent marker: Indicates semantic Agent
- "to" recipient: Indicates Goal/Recipient
- "from" source: Indicates Source Agent

4.2 Incorrect Semantic-Role Assignments

ERROR IDENTIFIED in Sentence 4:

"The contract was terminated by the company."

Given (Incorrect): Contract = Agent, Company = Object ✗

Correct: Contract = Patient (entity terminated), Company = Agent (performer) ✓

Analysis: System confused grammatical subject (Contract) with semantic agent. In passive voice, the subject is the UNDERGOER (patient), not the DOER (agent). The "by" phrase explicitly marks the true semantic agent.

4.3 Effects on Legal Information Extraction

Consequences of passive voice misinterpretation:

- "Obligation was breached" (no agent) → Cannot determine responsible party
- "Damages awarded to plaintiff" → Misidentifies who receives vs. who caused harm
- "Payment withheld by contractor" → May assign fault to Client instead
- "Insurance canceled by company" → Unclear who initiated; affects liability

CRITICAL: In legal contracts, identifying the Agent (who performed the action) determines liability and remedies. Confusing subject with agent results in incorrect breach attribution, reversed responsibility chains, and misallocated damages.

4.4 Recommended Syntax-Driven Architecture

Layered Semantic Role Labeling Pipeline:

LAYER 1: Deep Syntactic Parsing
- Dependency parsing (Enhanced Universal Dependencies)
- Identify: nsubj, nsubjpass (passive subject), agent (by-phrase), obl (oblique)
- Detect voice: active vs passive

LAYER 2: Semantic Role Labeling (SRL)
- PropBank/FrameNet with legal domain extensions
- Example: CAUSATION.TERMINATE frame
  ARG0 = Agent (terminator)
  ARG1 = Patient (terminated)
  ARG2 = Reason (why terminated)
- Neural SRL model (BERT-based) predicts roles

LAYER 3: Legal Domain Constraints
• Knowledge base: legal entities, capable roles, valid combinations
• Validate role assignments against domain knowledge
• Reason about causality: Who can legally terminate a contract?

LAYER 4: Linguistic Phenomena Handling
• Passive voice: if nsubjpass(X) & agent(Y) then Agent=Y, Patient=X
• Long-distance dependencies: track agent through relative clauses
• Prepositional phrases: "by company" overrides grammatical subject
• Implicit agents: flag ambiguous cases for review

LAYER 5: Extraction & Confidence Scoring
Output structured format with confidence scores
Confidence < 0.80 → Flag for human legal review
Confidence < 0.60 → Escalate to senior analyst

Key Technologies:

• Dependency Parser: Spacy, Stanford CoreNLP (legal domain)
• Semantic Role Labeler: AllenNLP, BERT-based models
• Legal Knowledge Base: Ontology of entities and valid role combinations
• Confidence Calibration: Ensemble of models; calibration on annotated corpus
• Interpretability: Provide explanation traces for review

Validation Strategy:

✓ Test on historical contracts with known outcomes
✓ Benchmark against human legal annotators
✓ Error analysis: Focus on passive, implicit agents, long-distance dependencies
✓ Continuous learning: Incorporate legal team feedback

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
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